Lista Derivadas

4.7

LISTA DERIVADAS

4.7) 4 5 6 8 12.

4) $Y = x^2 - 2x + 1 \quad y' = 2x - 1$

$Y(-2) = 2(-2) - 2 = -4 - 2 = \boxed{-6 = m}$

$y - y_0 = m(x - x_0) = (-2, 9),\ m\ 6$

$y - 9 = -6(x + 2)$

$y = -6x - 3$ // reta T

normal

$m = \frac{1}{m} = \frac{1}{-6} = \boxed{\frac{1}{6}}$

$y - 9 = 1/6(x + 2)$

$y = 1/6x + 2/6 + 9 = 1/6x + 55/6$

$y = \frac{1}{6}x + \frac{55}{6}$ //

5) $F(T) = 16T + T^2,\quad 0 \le T \le 8$

a) $Vm = \frac{F(b+H) - F(b)}{b + H - b} = \frac{F(b+H) - F(b)}{H}$

$F(16T + T^2) = 16(b + H) + (b + H)^2 = 16b + 16H + b^2 + 2bH + H^2$

$\frac{(16b + 16H + b^2 + 2bH + H^2) - (16b + b^2)}{H}$

$\frac{16H + 2bH + H^2}{H} = \boxed{16 + 2b + H}$

b) $(3, 3{,}1]$ $[3, 3{,}01]$, $[3, 3{,}001]$

1) 3; 3,1

$F(3) = 16(3) + 3^2 = 57$

$F(3{,}1) = 16(3{,}1) + (3{,}1)^2 = 59{,}21$

$Vm = \frac{59{,}21 - 57}{0{,}1} = 22{,}1$ //

2) 3; 3,01

$F(3{,}01) = 16(3{,}01) + (3{,}01)^2 = 57{,}2201$

$Vm = \frac{57{,}2201 - 57}{0{,}01} = \frac{0{,}2201}{0{,}01} = 22{,}01$ //

3) $F(3{,}001) = 16(3{,}001) + (3{,}001)^2 = 57{,}022001$

$Vm = \frac{57{,}022001 - 57}{0{,}001} = 22{,}001$

c) $F(T) = 16T + T^2 = F'(T) = 16 + 2T$ //

d) $T = 3 = 16 + 2(3) = 22$

6) $y(t) = \frac{b}{t} + ct$

a) $t=2$ $\quad v(t) = y'(t) = \frac{d}{dt}\left(\frac{b}{t} + ct\right) = \frac{b}{t^2} + c$

$y(2) = \frac{b}{2^2} + c = \frac{-b}{4} + c$

b) $a(t) = v'(t) = \frac{d}{dt}\left(\frac{b}{t^2} + c\right) = \frac{2b}{t^3}$

8(a) $1-4x^2$

$F(x+h) = 1-4(x+h)^2 =$

$1-4(x^2+2xh+h^2)$

$1-4x^2-8xh-4h^2$

$\lim_{h \to 0} \frac{F(x-h)-F(x)}{h}$

$\lim_{h \to 0} \frac{(1-4x^2-8xh-4h^2)-(1-4x^2)}{h}$

$\lim \left(\frac{-8xh-4h^2}{h}\right) = -8x-4h = -8x$

k) $\frac{d}{dx}(2x^2-3x-2) = 4x-3$

a) $f'(x) > 0$: $4x-3 > 0 = x > \frac{3}{4}$ // b) $4x-3 < 0 = x < \frac{3}{4}$ //

4.10

4,10) 5

5) $x=-2$

$\lim f(x) = 2-2(-2)^2 = 2-4 = -2$

Contínua em -2

Esquerda $= 2-x^2 \to f'(x) = -2x = -2(-2) = 4$

Direita $= -2 \to f'(x) = 0 = 0$

$\boxed{x=-2 \ \nexists}$

$x=2$ = Esquerda: $f(x) = -2 \Rightarrow f'(2) = 0$ /

Direita: $2x-6 = f'(x) = 2$ /

$x=2 \ \nexists$

4.12

4.12) 10 17 25 30

10) $(s^2-1)(3s-1)(5s^3+2s)$ | $f'(s) = u'vw + uv'w + uvw'$

$u =$ → $2s$

$v =$ → 3

w → $15s^2+2$

$(2s)(3s-1)(5s^3+2s) + (s^2-1)(3)(5s^3+2s) + (s^2-1)(3s-1)(15s^2+2)$

$\cdot(s^2-1)(3s-1)(5s^3+2s)$

17) $f(x) = 7(ax^2+bx+c)$

$f'(x) = 7.(2ax+b)$

25) $f(t) = 3t^3-4t+1$ | $f(0) = 3(0)^3-4(0)+1 = 1$

$f(0) - 7f'(0)$ | $f'(t) = \frac{d}{dt}(3t^3-4t+1) = 9t^2-4$

$f(0)-7f'(0) = 1-(7(-4)) = \boxed{1+47}$

30) $y = ax^2+bx$

$(1,5)$ → $y' = 2ax+b$

$m=8$ $(y'(1)=8)$ $x=1$ → $a(1)^2+b(1) = 5 = a+b=5$

$2a+b=8$ → a: $8-5 = a = 3$

$a+b=5$ → b: $3+b=5 = 2$

4.16

4.16) 3 9 13 24 55 65 83 88

3) $X(T) = 3T^2 - T^3$

a) $X(X)\ 3(0)^2 - (0)^3 = 0$ | $0 - 16 = -16\ m$

$(4)\ 3(4) - (4)^3 = 3(16) - 64 = -16$

b) $6T - 3T^2$

1 $6(1) - 3(1)^2 = 3$

2 $6(2) - 3(2)^2 = 0$

3 $6(3) - 3(3)^2 = -9$

4 $6(4) - 3(4)^2 = -24$

c) $6 - 6T$

1 $6 - 6 \cdot 1 = 0$

2 $6 - 6 \cdot 2 = -6$

3 $6 - 6 \cdot 3 = -12$

4 $6 - 6 \cdot 4 = -18$

9)

$\sqrt[3]{(3x^2+6x-2)^2} = (3x^2+6x-2)^{\frac{2}{3}}$

n da cadeia

$F(x) = [u(x)]^{\frac{2}{3}} \to \frac{2}{3} \cdot (3x^2+6x-2)^{-1/3} \cdot 6x+6$

$\to \frac{2(6x+6)}{3(3x^2+6x-2}$

55) $\frac{1}{2} \ln(7x^2-4)$

$u \to u' = 14x \Rightarrow \frac{1}{2} \cdot \frac{1}{7x^2-4} \cdot 14x$ | $f'(x) = \frac{14x}{2(7x^2-4)} = \frac{7x}{7x^2-4}$

83) $f(x) = e^{-x}$

$\to f(0) + x f'(0)$

$f(0) = e^{-0} = 1$

$f'(x) = \frac{d}{dx}(e^{-x}) = -e^{-x}$

$f'(0) = -e^{0} = -1$

$y = 1 + x(-1) = 1 - x$ //

88) $y = xe^{-x}$

$xy' = (1-x)y$

y') $y = x \cdot e^{-x} = y'\ 1 \cdot e^{-x} + (x(-e^{-x})) = e^{-x} - xe^{-x}$

xy') $x(e^{-x} - xe^{-x}) = xe^{-x} - x^2e^{-x}$

$(1-x)y = (1-x)(xe^{-x}) = xe^{-x} - x^2e^{-x}$

160 ats

14.21

4.21) 6 10 13

6) $y = e^{2x+1} = e \cdot e^{2x}$

0 $= y = e^{2x+1}$

1 $= 2e^{2x+1}$

2 $= 4e^{2x+1}$

3 $= 4e^{2x+1}$

10) $-2\cos\left(\frac{x}{2}\right)$ $u' = x/2 = 1/2$

$y' = -2 \cdot (-\text{sen}\, x/2) \cdot 1/x = \text{sen}\, x/2$

$y'' = \cos x/2 \cdot 1/2$

$y''' = -\text{sen}\, x/2 \cdot 1/2 \cdot 1/2 = \text{sen}\, x/2 \cdot 1/4$

$y^{IV} = -\cos x/2 \cdot 1/8$

$y^{V} = \text{sen}\, x/2 \cdot 1/16$

$\to \frac{1}{16} \cdot \left(\text{sen}\, \frac{x}{2}\right)$